

The Symmetric Group

These are a collection of notes on the symmetric group, taken from the book *The Representation Theory of the Symmetric Group* (by James & Kerber).

Let P be a permutation group on Ω and P' a permutation group on Ω' . Then we say P, P' are *similar* if there exists a bijection $\epsilon : \Omega \rightarrow \Omega'$ and an isomorphism $\phi : P \rightarrow P'$ so that for all $\pi \in P$ and $\omega \in \Omega$, we have $\phi(\pi)(\epsilon(\omega)) = \epsilon(\pi(\omega))$. Since P acts on Ω , we say $|\Omega|$ is the *degree of* P . Now we have the following result.

Proposition. Assume the above hypothesis. Then P, P' are similar iff they have the same degree.

Proof. In the forward direction, suppose P, P' are similar. By definition, there is a bijection $\epsilon : \Omega \rightarrow \Omega'$ (we may assume Ω, Ω' to be finite). Thus $|\Omega| = |\Omega'|$ and so P, P' have the same degree. Conversely, suppose they have the same degree. Say $\epsilon : \Omega \rightarrow \Omega'$ is a bijection. Put $\phi : P \rightarrow P'$ so that $\phi(\pi) = \epsilon \circ \pi \circ \epsilon^{-1}$ for $\pi \in P$. We have $\phi(\pi\tau) = \epsilon \circ \pi\tau \circ \epsilon^{-1}$, which equals $\phi(\pi)\phi(\tau)$, so ϕ is thus a homomorphism. If $\phi(\pi) = \phi(\tau)$, then $\epsilon \circ \pi\epsilon^{-1} = \epsilon \circ \tau\epsilon^{-1}$, so ϕ is 1-1. If $\tau \in P'$, then the element $\epsilon^{-1} \circ \tau \circ \epsilon \in P$ has image τ in P' , so ϕ is onto, and so ϕ is an isomorphism. Now fix $\pi \in P$ and $\omega \in \Omega$. We have $\phi(\pi) = \epsilon \circ \pi \circ \epsilon^{-1}$, so the left-hand side equals the right-hand side immediately, and we are done. $\square$

Lemma 1.1.7. We have $S_n = \langle (12), (23), \dots, (n-1\ n) \rangle = \langle (12), (1 \cdots n) \rangle$.

Proof. Suppose j, k in $\{1, \dots, n\}$ are distinct. WLOG, suppose $j < k$. We do strong induction on the difference $d = k - j > 0$. Then

$$(jk) = (k-2\ k) \cdots (j+1\ k-1)(j\ k-1)(k-1\ k) \cdots (j+1\ j+2)(j\ j+1)$$

and so by induction, (jk) is generated by $(12), \dots, (n-1\ n)$. Now fix $k \in \{2, \dots, n-1\}$. It suffices to that $(12), (1 \cdots n)$ generate $(k\ k+1)$. Notice that $(k\ k+1)$ equals $(1 \cdots n)^{-n+k-1}(12)(1 \cdots n)^{n-k+1}$, and so $(12), (1 \cdots n)$ generate $(k\ k+1)$, and we are done. $\square$

We define the *difference product* Δ_n to be 1 for $n = 1$ and $\Delta_n := \prod_{1 \leq i < j \leq n} (j - i)$ for $n \geq 2$. For $\pi \in S_n$, we naturally get an action by writing $\pi\Delta_n = \prod_{1 \leq i < j \leq n} (\pi(j) - \pi(i))$ and $\pi(-\Delta_n) = -\pi\Delta_n$. Since π is bijective, thus $\pi\Delta_n = \pm\Delta_n$. We define the *sign of* π by $\text{sgn} := \pi\Delta_n/\Delta_n$. Informally it can be said that $\text{sgn } \pi$ keeps track of the flips of the factors of Δ_n under the action by π .

Lemma 1.1.19. The map sgn (which takes a permutation π to $\text{sgn } \pi$) is a homomorphism $S_n \rightarrow \{1, -1\} \cong \mathbb{Z}/2\mathbb{Z}$. This map is onto iff $n \geq 2$.

Proof. Suppose $\pi, \tau \in S_n$. We know $\text{sgn}(\pi) = 1$ if π is even and -1 if π is odd. Suppose π can be decomposed into a transpositions and τ into b transpositions. Then $\text{sgn}(\pi\tau) = 1$ if $a + b$ is even and -1 if $a + b$ is odd. If a, b have the same parity, then $a + b$ is even, $\text{sgn}(\pi) = \text{sgn}(\tau)$, $\text{sgn}(\pi\tau) = 1$, and so $\text{sgn}(\pi)\text{sgn}(\tau) = 1 = \text{sgn}(\pi\tau)$.

If a, b do not have the same parity, then $a + b$ is odd, $\text{sgn}(\pi\tau) = -1$, and so $\text{sgn}(\pi)\text{sgn}(\tau) = -1 = \text{sgn}(\pi\tau)$ and so sgn is a homomorphism to the 2-element group. Suppose now $n \geq 2$. Taking $1 \in \{1, -1\}$, we have that the element $e \in S_n$ can be written in the form $(ab) \cdot (ab)$ (for some distinct $a, b \in \{1, \dots, n\}$), so hence $\text{sgn}(e) = 1$. Taking $-1 \in \{1, -1\}$, we have that any transposition in S_n is sent to -1 . So sgn is surjective. Conversely, suppose sgn is surjective. If instead $n = 0$ or $n = 1$, then we have a map from a one-element set to a two-element set, which cannot be surjective, and we are done. $\square$

Two permutations are conjugate iff they have the same cycle structure. We now introduce partitions. Suppose n is a positive integer. A *partition of n* is a sequence $\alpha = (\alpha_1, \alpha_2, \dots)$ of nonnegative integers α_i so that the sequence is weakly decreasing and $\sum_i \alpha_i = n$. Since $\alpha_i \geq 0$ for all i , there exists an index h so that $\alpha_j = 0$ for all $j > h$. Thus we can rewrite the partition $(\alpha_1, \alpha_2, \dots)$ as $(\alpha_1, \alpha_2, \dots, \alpha_h)$. Let $c(\pi)$ be the number of disjoint cyclic factors (including 1-cycles) of a permutation $\pi \in S_n$. Let $\alpha_i(\pi)$ denote the ordered lengths of the cyclic factors for $1 \leq i \leq c(\pi)$. Then we define the *cycle partition of π* to be the partition $(\alpha_1(\pi), \dots, \alpha_{c(\pi)}(\pi)) := \alpha(\pi)$. Let $a(\pi) := (a_1(\pi), \dots, a_n(\pi))$ denote the multiplicities of the parts of $\alpha(\pi)$. $a(\pi)$ is called the *cycle type of π* . We say an n -tuple $(a_1, \dots, a_n)$ is a *type of n* if for each $1 \leq i \leq n$, we have $a_i \in \mathbb{Z}_{\geq 0}$, and $\sum_i i a_i = n$. Let $C^S(\pi)$ denote the conjugacy class of the permutation $\pi \in S_n$. It follows that the elements of the same conjugacy class in S_n have the same cycle type and are the same types of n .

We now discuss Young subgroups of S_n . We say a partition $\lambda = (\lambda_1, \lambda_2, \dots)$ of n is *improper* if $\sum_i \lambda_i = n$. So the only difference between an improper partition and a partition as from before is that we do not require that $\lambda_1 \geq \lambda_2 \geq \dots$. Suppose $\lambda = (\lambda_1, \lambda_2, \dots)$ is an improper partition of n and let $\mathbf{n}_i^\lambda$ (for i at least 1) denote the subsets of $\mathbf{n} = \{1, \dots, n\}$, which are pairwise disjoint, and which satisfy $|\mathbf{n}_i^\lambda| = \lambda_i$ for all $i \geq 1$. Let S_i^λ denote the subgroup of S_n consisting of the only $\lambda_i!$ elements that leave each point of $\mathbf{n} \setminus \mathbf{n}_i^\lambda$ fixed. Then $S_\lambda := \prod_{i=1}^\infty S_i^\lambda$ is called a *Young subgroup* corresponding to the partition of n via $\lambda = (\lambda_1, \lambda_2, \dots)$. This Young subgroup is isomorphic to $S_{\lambda_1} \times S_{\lambda_2} \times \dots$ and is a subgroup of S_n . We also note that all subgroups S_λ that correspond to λ are conjugate to each other in S_n .

Suppose now F is a field. Then we get immediately two 1-dimensional representations of a Young subgroup S_λ , where λ is a partition of n . First, we get the trivial representation where each $\pi \in S_\lambda$ is sent to the identity element in $GL(F)$. Second, we get the sign representation, where each $\pi \in S_\lambda$ is sent to $e \in GL(F)$ (if π is even) and to $-e \in GL(F)$ (if π is odd). Let IS_λ and AS_λ denote the trivial and sign 1-dimensional representations of S_λ (respectively). Suppose μ is another improper partition of n . We now consider IS_λ , IS_μ , and AS_μ , induced up to S_n . An intertwining map between representations can be informally thought of as a map that commutes with homomorphisms as given by the representations. The space of intertwining maps has a dimension, which is called the intertwining number. By using Mackey's intertwining number theorem, $i(IS_\lambda \uparrow S_n, IS_\mu \uparrow S_n) = \sum_{S_\lambda \pi S_\mu} 1$. The index $S_\lambda \pi S_\mu$ is a double coset. So we conclude that the intertwining number of $IS_\lambda \uparrow S_n$ and $IS_\mu \uparrow S_n$ equals the number of double cosets $S_\lambda \pi S_\mu$ of S_λ and S_μ in

S_n , where $\pi \in S_n$. By more deductions, it follows that if $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_n)$ and $\mu = (\mu_1, \dots, \mu_n)$ are improper partitions of n with respective Young subgroups S_λ and S_μ , then the map $S_\lambda \pi S_\mu \rightarrow z_{ik} := |\mathbf{n}_i^\lambda \cap \pi[\mathbf{n}_k^\mu]|$ is a bijection between the set of double cosets of S_λ and S_μ in S_n and the set of $n \times n$ matrices (z_{ik}) over $\mathbb{Z}_{\geq 0}$ that satisfy $\sum_{i=1}^n z_{ik} \mu_k$ and $\sum_{k=1}^n z_{ik} \lambda_i$. From this we also have that the number of double cosets $S_\lambda \pi S_\mu$ is equal to the number of $n \times n$ matrices over the nonnegative integers with $\lambda = (\lambda_1, \dots, \lambda_n)$ as the row sums and $\mu = (\mu_1, \dots, \mu_n)$ as the column sums. Now take λ and μ as above. Then the number of matrices over the nonnegative integers with row sums λ_i and column sums μ_k equals the coefficient of $x^\lambda y^\mu = x_1^{\lambda_1} x_2^{\lambda_2} \dots x_n^{\lambda_n} y_1^{\mu_1} \dots y_n^{\mu_n}$ (written in multi-index notation) in the formal power series $\prod_{i,k \in \mathbb{N}} (1 - x_i y_k)^{-1}$. If we restrict the matrices to only have entries that are either 0 or 1, then we get the coefficient of $x^\lambda y^\mu$ in $\prod_{i,k} (1 + x_i y_k)$.

Now we explore pairs of partitions (α, β) of n so that the intertwining number $i(IS_\alpha \uparrow S_n, AS_\beta \uparrow S_n)$ equals 1. If $\text{char } F \nmid n!$, then there is an irreducible component of multiplicity 1 in both $IS_\alpha \uparrow S_n$ and $AS_\beta \uparrow S_n$. We may thus calculate the irreducible representations of S_n . Suppose we have a partition $\alpha = (\alpha_1, \dots, \alpha_h)$ of n . A *Young diagram* $[\alpha]$ of α can be visualized by placing the n nodes in h rows with α_i nodes in row i (here we assume the α_i are weakly decreasing). We get another partition of n by looking at the columns; this is called α' , the *partition associated with α* , and $[\alpha']$ is called the *Young diagram associated with $[\alpha]$* . For instance, if we have the Young diagram $[\alpha] = [(4, 2, 2, 1)]$:

$$\begin{array}{cccc} \times & \times & \times & \times \\ \times & \times & & \\ \times & \times & & \\ \times & & & \end{array}$$

then its associated Young diagram is $[(4, 3, 1, 1)]$ (also written as $[(4, 3, 1^2)]$). A Young diagram $[\alpha]$ that is equal to its associated Young diagram $[\alpha']$ is called *self-associated*. We define a partial order $\trianglelefteq$ on the set of partitions of n as follows. If α, β are partitions, then $\alpha \trianglelefteq \beta$ if for each i , we have $\sum_{k=1}^i \alpha_k \leq \sum_{k=1}^i \beta_k$, looking at the partial sums. Suppose now $\alpha \trianglelefteq \beta$. Then $\alpha \leq \beta$, where $\leq$ is the standard lexicographic order where $\alpha \leq \beta$ if $\alpha_i \leq \beta_i$ for all i . For if there is an i so that $\alpha_i > \beta_i$, then we get a contradiction by induction on i . For n , we say $(P(n), \wedge, \vee)$ is the *diagram lattice*, where $P(n)$ is the number of partitions of n , $\alpha \vee \beta$ is the supremum with respect to $\trianglelefteq$, and $\alpha \wedge \beta$ is the infimum with respect to $\trianglelefteq$. We now observe the following theorem.

Theorem (Gale-Ryser). If $\alpha = (\alpha_1, \alpha_2, \dots)$ and $\beta = (\beta_1, \beta_2, \dots)$ are partitions of n , then there exist matrices (with entries solely 0 or 1) with row sums $\alpha_1, \alpha_2, \dots$ and column sums $\beta'_1, \beta'_2, \dots$ iff $\alpha \trianglelefteq \beta$ (here the partition $(\beta'_1, \beta'_2, \dots)$ can be thought of as the column-wise Young diagram of $[\beta]$).

We have another result.

Theorem (Ruch-Schonhofer). If α, β are partitions of n , and S_α, S_β are Young subgroups of S_n that correspond to α and β' , and if the ground field F has characteristic not equal to 2, then the intertwining number $i(IS_\alpha \uparrow S_n, AS_{\beta'} \uparrow S_n) \neq 0$ iff $\alpha \trianglelefteq \beta$.

We now introduce the Young tableaux. Informally, a *Young tableau* t^α with Young diagram $[\alpha]$ (where α is a partition of n) can be thought of as replacing the nodes of $[\alpha]$ with the numbers $1, \dots, n$, each appearing exactly once.

We now consider the following theorem.

Theorem 2.1.3. Suppose α, α' are partitions of n , and S_α and $S_{\alpha'}$ are their corresponding Young subgroups. Then the induced representations $IS_\alpha \uparrow S_n$ and $AS_{\alpha'} \uparrow S_n$ share exactly one irreducible component, and with multiplicity 1. We write the irreducible component (as in the above theorem) as $[\alpha] = IS_\alpha \uparrow S_n \cap AS_{\alpha'} \uparrow S_n$. Subsequently, we have the following result.

Theorem 2.1.8 (Completeness). The set of $[\alpha]$, where α is a partition of n , is the set of ordinary irreducible representations of S_n .

As an example, we calculate the character table of S_4 . We know the number of conjugacy classes equals the number of irreducible representations. The conjugacy classes are indexed by cycle type (therefore, by partitions). In this case, the possible cycle types are a 1-cycle, 2-cycle, 2-2-cycle, 3-cycle, and a 4-cycle, so there are 5 irreducible representations. Immediately we have the trivial representation and the sign representation, where the sign representation sends odd permutations to -1 and even ones to 1. Both the trivial and sign representations are irreducible. We now look at the permutation character, which sends each permutation to the number of its fixed points. Maintaining the order of the conjugacy classes we had above, we get that the number of fixed points are 4,2,0,1,0. The norm of the permutation character is

$$\frac{1}{|S_4|}(4^2 \cdot 1 + 2^2 \cdot 6 + 1^2 \cdot 8) = 2 \neq 1$$

(where each addend is weighted by the number of elements in its conjugacy class), so the permutation character is not irreducible. Subtracting the trivial character from the permutation character yields the character (given by row entries) 3,1,-1,0,-1, and we get

$$\frac{1}{|S_4|}(3^2 \cdot 1 + 1^2 \cdot 6 + (-1)^2 \cdot 3 + (-1)^2 \cdot 6) = 1$$

so the permutation minus trivial character is irreducible. Tensoring the character in the previous sentence with the sign character yields another irreducible character, with row entries 3,-1,-1,0,1. By columnwise orthogonality of the character table, we get the last irreducible character of S_4 to have row entries 2,0,2,-1,0. So the character table of S_4 is given by

$$\begin{array}{ccccc} 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 & -1 \\ 3 & 1 & -1 & 0 & -1 \\ 3 & -1 & -1 & 0 & 1 \\ 2 & 0 & 2 & -1 & 0 \end{array}$$

Suppose now we have partitions $\alpha^1, \dots, \alpha^{P(n)}$ that are lexicographically ordered, i.e. $\alpha^1 = (1^n), \dots, \alpha^{P(n)} = (n)$. Define a multiplicity matrix M_n so that the entries are

$m_{ik} := (IS_{\alpha_i} \uparrow, [\alpha]^k)$. It then follows that M_n is an upper-triangular matrix with 1s on the diagonal.

Suppose now α, β are partitions of n . Define the *outer product* of $[\alpha]$ and $[\beta]$ to be $[\alpha][\beta] := [\alpha] \# [\beta] \uparrow S_{m+n}$, where $[\alpha] \# [\beta]$ is the ordinary irreducible representation (induced) of $S_m \times S_n$. The outer product is associative and commutative. For instance, we have $IS_\alpha \uparrow S_n = [\alpha_1][\alpha_2] \cdots [\alpha_h]$.

We define the *Jacobi-Trudi determinant* for $\alpha = (\alpha_1, \dots, \alpha_h)$ to be $\xi^\alpha = ||[\alpha_i + j - i]||_{1 \leq i, j \leq h}$ with $[r] = [r]$ if $r > 0$, $[0] = 1$, and $[r] = 0$ if $r < 0$. Suppose now π is a permutation with a k -cycle. Thne the Murnaghan-Nakayama rule states that $\xi^\alpha(\pi) = \sum_{\text{rim } k\text{-hooks in } R} (-1)^{h(R)-1} \xi^{\alpha \setminus R}(\pi')$, where a rim k -hook (also known as a ribbon / border-strip) is a connected set of k -cells within a Young diagram that can be removed to leave behind a smaller partition μ , $h(R)$ equals the hook length of rim k -hook R , and π' is π without the k -cycle. We now note the following theorem.

Theorem 2.4.2 (Branching). Suppose $\alpha = (\alpha_1, \alpha_2, \dots)$ is a partition of n . Then we have for the restriction of $[\alpha]$ to the stabilizer S_{n-1} of the point $[\alpha] \downarrow S_{n-1} = \sum_{i: \alpha_i > \alpha_{i+1}} [\alpha^{i-}]$. On the other hand, if S_n denotes the stabilizer of the point $n+1$ in S_{n+1} , then we have for the induced representation $[\alpha] \uparrow S_{n+1} = \sum_{i: \alpha_i < \alpha_{i+1}} [\alpha^{i+}]$.

As an example, take the partition $\alpha = (3, 2, 1, 1)$ of $n = 7$. If we remove a box from row 1 of $[\alpha]$, then we get $(2, 2, 1, 1)$, which is a valid partition. If we remove a box from row 2, we get $(3, 1^3)$, which is also valid. If we remove a box from row 3, we get $3, 2, 0, 1$, which is not a valid partition. If we remove a box from row 4, we get $(3, 2, 1, 0)$, which is valid. So $[3, 2, 1^2] \downarrow S_6 = [2^2, 1^2] + [3, 1^3] + [3, 2, 1]$. A similar logic applies for “partitions induced to symmetric groups on a higher number of elements.”

We have the following theorem.

Theorem 2.5.7. Suppose α is a partition of $n > 1$. Then:

1. if $\alpha \neq \alpha'$, then $[\alpha] \downarrow A_n = [\alpha'] \downarrow A_n$ is irreducible, while
2. if $\alpha = \alpha'$, then $[\alpha] \downarrow A_n = [\alpha'] \downarrow A_n$ splits into two irreducible and conjugate representations $[\alpha]^\pm$ of A_n , i.e. $\alpha^{+(12)}$, defined by $[\alpha]^{+(12)}((12)\pi(12)) := [\alpha]^+\pi$ (for $\pi \in A_n$) is equivalent to $[\alpha]^-$. A complete system of equivalence classes of ordinary irreducible representations of A_n is therefore $\{[\alpha] \downarrow A_n \mid \alpha \neq \alpha'\} \cup \{[\alpha]^\pm \mid \alpha = \alpha' \vdash n\}$.

We now note the following theorem.

Theorem 2.6.1. If a finite group G has the same character table as S_n , then $G \cong S_n$.

A *rim q -hook* is a connected segment of the diagram boundary of length q . A *q -core* is a diagram with no rim q -hook.

Theorem 2.7.16. Every partition α has a uniquely determined q -core $[\bar{\alpha}]$, obtained by successive rim q -hook removal.

We have the following theorem.

Theorem (Young's Rule). Suppose β, γ are partitions of n . Then we have a decomposition of the outer product $[\beta][\gamma]$ as $\sum_v K_{\beta, \gamma}^v[v]$, where $K_{\beta, \gamma}^v[v]$ counts the semistandard tableaux of shape v/β and content γ .

We now give a few related definitions. A *standard Young tableau* is a Young tableau (for n) so that every integer $1, \dots, n$ appears exactly once and the numbers in rows and columns are strictly-increasing (from left to right and top to bottom, respectively). A *semistandard Young tableau (SSYT)* is a Young tableau (for n) so that the numbers in the rows are weakly increasing (from left to right) and the numbers in the columns are strictly-increasing (from top to bottom). If the diagram $[\beta]$ fits inside the diagram $[v]$, then by *shape* v/β , we mean: take $[v]$ and delete the boxes that correspond to $[\beta]$. Suppose now we have a tableau T . The *content* for T is a sequence $\gamma = (\gamma_1, \gamma_2, \dots)$ so that γ_1 equals the number of 1s in T , γ_2 equals the number of 2s in T , etc.

Suppose now T is a tableau. Reading T from left to right and top to bottom, if the sequence of entries i is so that at each point $j \in \{1, \dots, n\}$, the number of $(j-1)$ s seen so far is at least the number of j s, then the sequence of i s is called a *lattice partition*. We have the following result.

Littlewood-Richardson Rule. Suppose $\alpha \vdash m$, $\beta \vdash n$, and $\gamma \vdash m+n$. Then the multiplicity $([\alpha][\beta], [\gamma])$ equals the number of SSYT of shape γ/α and content β , which yield lattice permutations when reading their entries left to right and downwards (like a book).

We now discuss an example. Suppose $\alpha = (3, 2)$ and $\beta = (2, 1)$. We want to find $[\alpha][\beta]$. In case 1, we can put the 1s in the first row and the 2 in the second row, giving $[5, 3]$. In case 2, we can put the 1s in the first row and the 2 in a third row, giving $[5, 2, 1]$. In case 3, we can put one 1 in the first row and the other 1 and 2 in the second row, giving $[4, 4]$. In case 4, we can either put the 1st in the first and second rows and the 2 in a third row or 1 in the first with 2 in the second row with 1 in a third row; both of these give $[4, 3, 1]$. In case 5, we can put one 1 in the first row and make a third row with 1 2, giving $[4, 2, 2]$. In case 6, we can put one 1 in the first row, one 1 in a third row, and the 2 in a fourth row, giving $[4, 2, 1, 1]$. In case 7, we can put one 1 in the second row, and make a third row with 1 2, giving $[3, 3, 2]$. In case 8, we can put the 1s in a third row and the 2 in a fourth row, giving $[3, 2, 2, 1]$. In case 9 (the last case), we can put one 1 in row 2, one 1 in a third row, and a 2 in the fourth row, giving $[3, 3, 1, 1]$. So we get a decomposition of $[\alpha][\beta]$ into $[5, 3] + [5, 2, 1] + [4, 4] + 2[4, 3, 1] + [4, 2, 2] + [4, 2, 1, 1] + [3, 3, 2] + [3, 2, 2, 1] + [3, 3, 1, 1]$.

We can complete $[\alpha] \otimes [\beta]$ (for partitions α, β) by first using the determinantal formula and $\xi^\alpha \otimes \xi^\beta$, then decomposing via the Littlewood-Richardson rule on the induced

characters.

Lemma 2.9.16. $\xi^\alpha \otimes \xi^\beta = \sum_v \xi^v$, if the sum is taken over all the $k \times k$ matrices $v = (v_{ij})$ such that $v_{ij} \in \mathbb{N}$ and $\sum_i v_{ij} = \beta_j$ and $\sum_j v_{ij} \alpha_i$.

Now we discuss wreath products. Suppose G is a group and H is a subgroup of S_n . Let $G^n = \{\text{maps } f \mid f : \mathbf{n} \rightarrow G\}$. Then the *wreath product* $G \wr H$ is defined to be $G^n \rtimes H = \{(f; \pi) \mid f : \mathbf{n} \rightarrow G, \pi \in H\}$. We get a multiplication on $G \wr H$ by $(f; \pi) \cdot (f'; \pi') := (ff'; \pi\pi')$, where $f_\pi := f \circ \pi^{-1}$, and $(ff')(i) := f(i)f'(i)$ for $i \in \{1, \dots, n\}$. The order of the wreath product $G \wr H$ is $|G|^n \cdot |H|$.

We now give a few examples of wreath products. For the group $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, the group $\mathbb{Z}/2\mathbb{Z}$ acts on it by swapping elements from the copies of $\mathbb{Z}/3\mathbb{Z}$. We have also B_n , the group of signed permutations, which is the two-element group $\{\pm 1\}$ acting on S_n by sending even permutations to themselves and odd permutations to minus themselves. As an unusual example, consider an n -gon with a choice of two colors for each of the n vertices. The symmetry group of the n -gon is D_{2n} , but we have also $\mathbb{Z}/2\mathbb{Z}$ acting on each vertex by dictating the appearing color (say one of the colors is the “identity color”); this gives the wreath product $\mathbb{Z}/2\mathbb{Z} \wr D_{2n}$.

We now discuss the conjugacy classes of $G \wr S_n$. Suppose $(f; \pi) \in G \wr S_n$ and $\pi = (i_1 i_2 \dots i_r)$. Then the *cycle product* is defined to be $g_j(f; \pi) := f(i_r)f(i_{r-1}) \dots f(i_1) \in G$. These cycle products are conjugacy-class-invariant. Suppose G has k conjugacy classes. Then the *type* of $(f; \pi)$ gives, for each conjugacy class $C^{(i)}$ and length j , the number of cycles of π of length j that have cycle product in $C^{(i)}$. We use the notation $a(f; \pi)$ to denote (a_{ik}) , where a_{ik} is the number of cycles of π (with length j) with cycle product in $C^{(i)}$. We define a *k-multipartition* of n to be $\lambda = (\lambda^{(1)}, \dots, \lambda^{(k)})$, where each $\lambda^{(i)}$ is a partition and $\sum_i |\lambda^{(i)}| = n$.

Theorem 4.2.8. Two elements $(f; \pi)$ and $(f'; \pi')$ of $G \wr S_n$ are conjugate iff $a(f; \pi) = a(f'; \pi')$.

Now we look at representations of wreath products over algebraically closed fields. Let $G^* := \{(f; \text{id}_n) \mid f \in G^n\} \cong G^n \trianglelefteq G \wr H$. Let $(\#D)^{\otimes r}$ denote the module $D \otimes \dots \otimes D$ (with r factors) with G^* acting on the r components. If $\alpha = (\alpha_1, \dots, \alpha_h)$ is a partition of n and $D_1, \dots, D_h$ are irreducibles of G , then we get a G^* -module $D_1^{\otimes \alpha_1} \otimes \dots \otimes D_h^{\otimes \alpha_h}$.

Suppose now $G = S_m$ and $H = S_n$. Then the irreducible representations of the wreath product $S_m \wr S_n$ are indexed by $P(m)$ -multipartitions of n (where $P(m)$ is the number of partitions of m). Then we get the *wreath product Specht module* to be

$$S_v = \text{Ind}[S_{\mu_1}^{\otimes |v^{(1)}|} \otimes \dots \otimes S_{\mu_r}^{\otimes |v^{(r)}|} \otimes S_{v^{(1)}} \otimes \dots \otimes S_{v^{(r)}}]$$

where $\mu = (\mu_1, \dots, \mu_r)$ is a partition of m , with the μ_i strictly-decreasing.